

Swiss Dwellings: A large dataset of apartment models including aggregated geolocation-based simulation results covering viewshed, natural light, traffic noise, centrality and geometric analysis

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Description

Introduction

This dataset contains detailed data on 42,207 apartments (242,257 rooms) in 3,093 buildings including their geometries, room typology as well as their visual, acoustical, topological and daylight characteristics.

Procurement

The data is sourced from commercial clients of [Archilyse AG](#) specializing on the digitization and analysis of buildings. The existing building plans of clients are converted into a geo-referenced, semantically annotated representation and undergo a manual Q/A process to ensure accuracy of the data and to ensure a maximum 5%-deviation in the apartments' areas (validated with a median deviation of 1.2%).

Geometries

The dataset contains a file `geometries.csv` which contains the geometries of all areas, walls, railings, columns, windows, doors and features (sinks, bathtubs, etc.) of an apartment.

In total the datasets contains the 2D geometry of ~1.2 million separators (walls, railings), ~550,000 openings (windows, doors), ca. 400,000 areas (rooms, bathrooms, kitchens, etc.) and ~240,000 features (sinks, toilets, bathtubs, etc.).

Each row contains:

- `entity_type`: The entity type (*area, separator, opening, feature*)
- `entity_subtype`: The entity's sub type (e.g. *WALL*)
- `geometry`: The element's geometry as a [WKT](#) geometry. The geometry is given in the site's local coordinate system. I.e. the position between elements of the same site are correct in respect to each other. The +y direction points northwards, the +x direction points eastwards.
- `area_id`: The ID of the area in which the element is spatially contained (for features)
- `unit_id`: The ID of the unit in which the element is spatially contained (for features, areas)
- `apartment_id`: The ID of the apartment (for features, areas)
- `floor_id`: The ID of the floor
- `building_id`: The ID of the building
- `site_id`: The ID of the site

An example:

column

<code>entity_type</code>	area
<code>entity_subtype</code>	ROOM
<code>geometry</code>	POLYGON ((-2.10406 4.02039...
<code>site_id</code>	127
<code>building_id</code>	164
<code>floor_id</code>	12864
<code>apartment_id</code>	d4438f2129b30290845ce7eef98a5ba7

column

unit_id	76643
area_id	684674

Simulations

Beside the geometrical model, we also provide simulation data on the visual, acoustic, solar, layout and connectivity-related characteristics of the apartments. The file `simulations.csv` contains the simulation data aggregated on a per-area basis. Each row contains the identifier

columns `area_id`, `unit_id`, `apartment_id`, `floor_id`, `building_id`, `site_id` as defined above as well as 367 simulation columns. Each simulation column is formatted as:

`<simulation_category>_<simulation_dimensions>_<aggregation_function>`

For instance, the column `view_buildings_median` describes the amount of building surface that can be seen from any point in a given room. The aggregation methods vary per simulation category and are described in detail below.

Layout

The *layout* features represent simple features based on the geometry and composition of a room, the dataset provides the following information in an unaggregated form.

Area Basics / Geometry

dimension	description
layout_area_type	The area's area type
layout_net_area	The area's share of the apartment's net area (e.g. 0 for a balcony)
layout_area	The area's actual area
layout_perimeter	The area's perimeter
layout_compactness	The area's compactness (the Polsby–Popper score)
layout_room_count	The area's share to the apartment's room count
layout_is_navigable	True if the area is navigable by a wheelchair

Area Features

dimension	description
layout_has_sink	True if the area has a sink
layout_has_shower	True if the area has a shower
layout_has_bathtub	True if the area has a bathtub
layout_has_toilet	True if the area has a toilet
layout_has_stairs	True if the area has stairs
layout_has_entrance_door	True if the area is directly leading to an exit of the apartment

Area Windows / Doors

dimension	description
layout_number_of_doors	The number of doors directly leading to the area
layout_number_of_windows	The number of windows of the area
layout_door_perimeter	The sum of all door lengths directly leading to the area
layout_window_perimeter	The sum of all window lengths of the area

Area Walls / Railings

dimension	description
layout_open_perimeter	The sum of all of the areas boundaries that are neither walls nor railings
layout_railing_perimeter	The sum of all of the areas boundaries that are railings
layout_mean_walllengths	The mean length of the area's sides
layout_std_walllengths	The standard deviation of the lengths of the area's sides

Area Adjacency

dimension	description
layout_connects_to_bathroom	True if the area connects to a bathroom
layout_connects_to_private_outdoor	True if the area connects to an outside area that is private to the apartment

View

The views from an object help to understand the impact of the surroundings on the object. The view simulation calculates the visible amount of buildings, greenery, water etc. on each individual hexagon from the analyzed object. The values are expressed in steradians (sr) and represent the amount a certain object category occupies in the spherical field of view.

Each of the following dimension is provided using the room-wise aggregations *min*, *max*, *mean*, *std*, *median*, *p20* and *p80*. For instance, the column `view_greenery_p20` describes the amount of greenery that can be seen from at least 20% of the positions in the area.

dimension	description
view_buildings	The amount of visible buildings
view_greenery	The amount of visible greenery
view_ground	The amount of visible ground
view_isovist	The amount of visible isovist
view_mountains_class_2	The amount of visible mountains of UN mountain class 2
view_mountains_class_3	The amount of visible mountains of UN mountain class 3
view_mountains_class_4	The amount of visible mountains of UN mountain class 4
view_mountains_class_5	The amount of visible mountains of UN mountain class 5
view_mountains_class_6	The amount of visible mountains of UN mountain class 6
view_railway_tracks	The amount of visible railway_tracks
view_site	The amount of visible site

dimension	description
view_sky	The amount of visible sky
view_tertiary_streets	The amount of visible tertiary_streets
view_secondary_streets	The amount of visible secondary_streets
view_primary_streets	The amount of visible primary_streets
view_pedestrians	The amount of visible pedestrians
view_highways	The amount of visible highways
view_water	The amount of visible water

Sun

Sun simulations help to understand the impact of the solar radiation on the object. The outcome of the sun simulations helps to identify surfaces that have great solar potential. Sun simulations are defined by the amount of sun radiation on each individual hexagon from the analyzed object. The sun simulation not only includes direct sun but also considers scattered light. The sun simulation values are given in Kilolux (klx). Simulations are performed for the days of summer solstice, winter solstice and vernal equinox.

Each of the following dimension is provided using the room-wise aggregations *min*, *max*, *mean*, *std*, *median*, *p20* and *p80*. For instance, column `sun_201806211200_median` describes the median amount of direct daylight received on the positions in the area.

Vernal Equinox

dimension	description
sun_201803210800	Daylight at 08:00 on 21st of March
sun_201803211000	Daylight at 10:00 on 21st of March
sun_201803211200	Daylight at 12:00 on 21st of March
sun_201803211400	Daylight at 14:00 on 21st of March
sun_201803211600	Daylight at 16:00 on 21st of March

dimension	description
sun_201803211800	Daylight at 18:00 on 21st of March

Summer Solstice

dimension	description
sun_201806210600	Daylight at 06:00 on 21st of June
sun_201806210800	Daylight at 08:00 on 21st of June
sun_201806211000	Daylight at 10:00 on 21st of June
sun_201806211200	Daylight at 12:00 on 21st of June
sun_201806211400	Daylight at 14:00 on 21st of June
sun_201806211600	Daylight at 16:00 on 21st of June
sun_201806211800	Daylight at 18:00 on 21st of June
sun_201806212000	Daylight at 20:00 on 21st of June

Winter Solstice

dimension	description
sun_201812211000	Daylight at 10:00 on 21st of December
sun_201812211200	Daylight at 12:00 on 21st of December
sun_201812211400	Daylight at 14:00 on 21st of December
sun_201812211600	Daylight at 16:00 on 21st of December

Noise / Window Noise

Noise level and the distribution of elements from an area helps to understand how an object is exposed to the acoustics of this area. The acoustic simulation calculates the noise intensity on each individual hexagon from the analyzed object considering traffic and train noise datasets. Adjacent buildings are considered as noise blocking elements. The values are expressed in dBA (decibels).

Window Noise

The noise per window of a given area is aggregated via min and max. For instance, `window_noise_train_day_max` represents the maximum amount of noise received on any window of the area.

dimension	description
<code>window_noise_traffic_day</code>	The amount of noise received on the area's windows from daytime car traffic
<code>window_noise_traffic_night</code>	The amount of noise received on the area's windows from night-time car traffic
<code>window_noise_train_day</code>	The amount of noise received on the area's windows from daytime train traffic
<code>window_noise_train_night</code>	The amount of noise received on the area's windows from night-time train traffic

Area-Wise Noise

The area-wise noise describes the amount of noise received from a noise source aggregated over the whole area in an unaggregated form. For instance, `noise_traffic_night` describes the dBA of noise received in the area from car traffic at night when propagating noise from all windows.

dimension	description
<code>noise_traffic_day</code>	The amount of noise received in the area from daytime car traffic
<code>noise_traffic_night</code>	The amount of noise received in the area from night-time car traffic
<code>noise_train_day</code>	The amount of noise received in the area from daytime train traffic
<code>noise_train_night</code>	The amount of noise received in the area from night-time train traffic

Connectivity

Centrality simulations help to analyze a floor plan, whether it's a shopping mall and you want to identify prominent areas in order to select the most prominent spot or it's an interior design circulation path and you want to determine open floor plan areas. Centrality simulations are

done using topological measures that score grid cells by their importance as a part of a gridcell network.

The distances and centralities are aggregated via *min*, *max*, *mean*, *std*, *median*, *p20* and *p80*. For instance, `connectivity_balcony_distance_min` describes the shortest distance to the next balcony from the point closest to the balcony in the area.

Distances

dimension	description
<code>connectivity_room_distance</code>	Distance to the next area of type ROOM
<code>connectivity_living_dining_distance</code>	Distance to the next area of type LIVING_DINING
<code>connectivity_bathroom_distance</code>	Distance to the next area of type BATHROOM
<code>connectivity_kitchen_distance</code>	Distance to the next area of type KITCHEN
<code>connectivity_balcony_distance</code>	Distance to the next area of type BALCONY
<code>connectivity_loggia_distance</code>	Distance to the next area of type LOGGIA
<code>connectivity_entrance_door_distance</code>	Distance to the next apartment exit

Centralities

dimension	description
<code>connectivity_eigen_centrality</code>	The Eigen-Centrality value
<code>connectivity_betweenness_centrality</code>	The Betweenness-Centrality value
<code>connectivity_closeness_centrality</code>	The Closeness-Centrality value

Files

geometries.csv

Files (2.3 GB)

Additional details

Related works

Citations to this version

Page size:

10